Week 5

Abstract Class, Interface and Static Members

Program 13

Write a Java program demonstrating an abstract class and abstract method.

```
abstract class Shape {
   abstract void display();
}

class Circle extends Shape {
   void display() {
      System.out.println("This is a Circle");
   }

   public static void main(String[] args) {
      Circle c = new Circle();
      c.display();
   }
}
```

OUTPUT:

This is a Circle

Program 14

Write a Java program demonstrating interface implementation.

```
interface Printable {
   void print();
}

class Student implements Printable {
   public void print() {
      System.out.println("Student details are displayed");
   }

   public static void main(String[] args) {
      Student s = new Student();
      s.print();
   }
}
```

OUTPUT:

Student details are displayed

Program 15

Write a Java program illustrating the use of static members, final keyword, and super keyword.

```
class Person {
```

```
    String name = "Rahul";

    final int age = 20;

    static String college = "Amity University";

    void display() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("College: " + college);
    }
}

class Student extends Person {
    String name = "Amit";

    void show() {
        System.out.println("Student Name: " + name);
        System.out.println("Person Name: " + super.name);
        System.out.println("Age: " + super.age);
        System.out.println("College: " + Person.college);
    }

    public static void main(String[] args) {
        Student s = new Student();
        s.show();
    }
}
```

OUTPUT:
Student Name: Amit
Person Name: Rahul
Age: 20
College: Amity University